AyuAware Nutrition Section

Technical Product Requirements Document

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1. Executive Summary

What We're Building

The Nutrition section transforms meal tracking from calorie counting into **Ayurvedic nutrition guidance**. It answers the fundamental question: "*Is what I'm eating supporting or disrupting my unique constitution?*"

This isn't a typical food diary. It's a personalized nutrition coach that evaluates food through the lens of:

Dimension	What We Measure	Why It Matters
Agni Compatibility	Can your digestive fire handle this food?	Poor Agni = poor nutrition absorption
Dosha Impact	Does this food balance or aggravate doshas?	Food directly affects Vata/Pitta/Kapha
Food Qualities	Hot/cold, heavy/light, dry/oily	Qualities should oppose current imbalances
Meal Timing	When are you eating relative to Agni	Timing determines digestion quality
Mindful Eating	How are you eating?	Rushed eating = compromised

Why It Matters

Most nutrition apps track calories, macros, and micronutrients. But Ayurveda teaches that:

- **The same food affects different people differently** — A salad might be perfect for Pitta but aggravate Vata
- **When you eat matters as much as what you eat** — Heavy meals at night overwhelm weakened Agni
- **How you eat transforms digestion** — Rushed eating creates Ama (toxins) regardless of food quality
- **Food is medicine** — The right foods can correct imbalances; the wrong ones create them

Key Differentiator

Standard food tracking: "You ate 2,100 calories, 45g protein"

AyuAware Nutrition: "Your lunch was Pitta-aggravating due to spice and salt. Your Agni score of 68% struggled with the heavy ramen (compatibility: 46/100). Consider cooling, lighter dinner options like kitchari or steamed vegetables. Eat before 7 PM while Agni is still active."

Every meal is analyzed for **personal impact**, not generic nutritional content.

2. Section Overview

2.1 Information Architecture

```
Nutrition Section
│
├── 🍽️ Today's Nutrition Overview
│    ├── Meals logged (breakfast/lunch/dinner/snacks)
│    ├── Dosha balance from food
│    ├── Agni compatibility score
│    ├── Eating pattern insights
│    └── Dinner recommendations
│
├── ➕ Quick Log Meal (Floating Action Button)
│    ├── Photo capture with AI recognition
│    ├── Manual entry with search
│    ├── Recent meals
│    └── Favorites
│
├── 🥗 Constitutional Food Guide
│    ├── Foods for your Prakriti
│    ├── Current recommendations (Vikriti)
│    ├── Seasonal adjustments
│    └── Weather-based suggestions
│
├── ⏰ Meal Timing & Agni Optimization
```

```
│       ├── Optimal eating windows
│       ├── Agni strength curve
│       ├── Meal spacing guidance
│       └── Mindful eating practices
│
├── 📊 Weekly Nutrition Overview
│       ├── Meal logging consistency
│       ├── Agni compatibility trends
│       ├── Dosha balance trends
│       ├── Taste balance analysis
│       └── Recommendations
│
└── 📋 Meal Plans & Recipes
        ├── Dosha-balancing meal plans
        ├── Constitutional recipes
        ├── Shop integration
        └── Custom plan creation
```

2.2 The Ayurvedic Nutrition Philosophy

Food as Medicine

In Ayurveda, food isn't just fuel — it's the primary medicine. The right foods can:

- Correct dosha imbalances
- Strengthen Agni (digestive fire)
- Build Ojas (vitality)
- Clear Ama (toxins)

The 20 Food Qualities (Gunas)

Ayurveda categorizes food by qualities, not just nutrients:

Quality Pair	Vata Effect	Pitta Effect	Kapha Effect
Hot / Cold	Hot pacifies	Cold pacifies	Hot pacifies
Heavy / Light	Heavy pacifies	Light pacifies	Light pacifies
Oily / Dry	Oily pacifies	Dry aggravates	Dry pacifies
Smooth / Rough	Smooth pacifies	Smooth neutral	Rough pacifies

The Six Tastes

Every meal should ideally include all six tastes:

Taste	Sanskrit	Effect	Examples
Sweet	Madhura	Nourishing, grounding	Grains, dairy, sweet fruits

Sour	Amla	Stimulating, heating	Citrus, fermented foods
Salty	Lavana	Moistening, heating	Salt, seaweed
Pungent	Katu	Stimulating, drying	Spices, onion, garlic
Bitter	Tikta	Cooling, detoxifying	Leafy greens, turmeric
Astringent	Kashaya	Drying, cooling	Beans, unripe fruits

2.3 Design Philosophy

Agni-Centric Evaluation Every food is evaluated against the user's current Agni strength. A food might be constitutionally appropriate but still problematic if Agni can't digest it.

Constitution-First Recommendations Generic "healthy eating" advice is replaced with personalized guidance based on Prakriti and current Vikriti.

Timing as Critical Factor When you eat determines how well you digest. The section emphasizes optimal meal windows aligned with the natural Agni cycle.

Mindful Eating Integration How you eat matters. We track eating conditions (rushed vs. calm, distracted vs. focused) as factors in digestion quality.

Action-Oriented Insights Every analysis ends with specific recommendations. Not just "your meal was heavy" but "take a 10-minute walk and have ginger tea."

3. User Journey Breakdown

3.1 First-Time Visitor Experience

Context: User navigates to Nutrition section for the first time.

Twin Message:

"Welcome to Nutrition! Let me teach you about eating for your unique constitution..."

Educational Card:

```
┌──────────────────────────────────────────────┐
│ 💡 Ayurvedic Nutrition                       │
│ ──────────────────────────────────────────── │
│                                              │
│ In Ayurveda, food is medicine. What you eat  │
│ should:                                      │
│                                              │
│ ✓ Match your constitution (Prakriti)         │
│ ✓ Balance current imbalances (Vikriti)       │
│ ✓ Support your Agni (digestive fire)         │
│ ✓ Suit the season and weather                │
```

```
It's not about calories or macros — it's
about food qualities:

• Hot vs. Cold
• Heavy vs. Light
• Dry vs. Oily
• Rough vs. Smooth

I'll guide you toward foods that support
YOUR unique balance.

              [Got it! →]
```

After user taps "Got it!", section loads with Today's Overview.

3.2 Returning User Experience

Context: User returns to Nutrition section.

Twin Message:

"Let's see what you've eaten today and how it's supporting your balance..."

Immediately shows Today's Nutrition Overview card.

3.3 Viewing Today's Nutrition

Context: User views the daily nutrition overview.

Collapsed Today's Nutrition Card:

```
🍽️ Today's Nutrition
──────────────────────────────────────

Meals Logged: 2 of 3

[Breakfast] ✓  [Lunch] ✓  [Dinner] ○

Dosha Balance from Food:
[Small Tri-Color Chart]
Slightly Pitta-aggravating

Agni Compatibility: 78/100
Most foods suit your digestive fire
```

```
│                                                │
│ ⚠️ Lunch was heavy — gentle dinner recommended │
│                                                │
│            [See Full Details ↓]                │
└────────────────────────────────────────────────┘
```

Key Elements:

- **Meal status row** — Visual checkmarks for logged meals, circles for pending
- **Dosha balance chart** — Small tri-color donut showing food's dosha impact
- **Agni compatibility score** — 0-100 scale showing digestive compatibility
- **Alert/insight** — Context-aware recommendation

Expanded Today's Nutrition Card:

1. **Date Header**
   - "Saturday, January 31"
2. **Meals Today** (Individual meal cards)
   - Each meal shows: thumbnail, name, time, Agni score, dosha impact
   - Tap to view full analysis
   - "+ Log Dinner" button for pending meals
3. **Overall Dosha Balance**
   - Larger tri-color chart
   - Percentage breakdown: Vata 35%, Pitta 45%, Kapha 20%
   - Insight: "Today's food is slightly Pitta-aggravating..."
4. **Agni Compatibility Analysis**
   - Score: 78/100
   - Explanation of which meals helped/challenged
   - Best meal: Breakfast (85/100)
   - Challenging meal: Lunch (46/100)
5. **Eating Patterns**
   - Meal timing assessment
   - Meal spacing check
   - Mindful eating scores per meal
6. **Recommendations for Dinner**
   - Specific suggestions based on what's been eaten
   - Recipe links
   - Action buttons: [See Suggested Recipes →] [Log Dinner Now →]

3.4 Logging a Meal with Photo

Context: User taps floating "+ LOG MEAL" button.

Step 1: Meal Selection Modal

```
┌────────────────────────────────────────────────┐
│ Log a Meal                                  [×]│
```

```
Which meal?
[☀️ Breakfast] [☀️ Lunch] [🌙 Dinner] [☕ Snack]

How would you like to log it?

📷 Take Photo
AI will recognize your food

✏️ Manual Entry
Search or type food items

🕐 Recent & Favorites
Quickly log repeated meals
```

Step 2: Camera Capture

- Full-screen camera viewfinder
- Guidelines overlay for food placement
- Tips: "Good lighting, top-down view"
- Buttons: [Capture] [Cancel] [Gallery]

Step 3: AI Processing

Twin Message:

"Let me analyze your food..." [Spinner animation]

Step 4: Recognition Results

```
[Food Photo]

I see:
✓ Ramen noodles
✓ Boiled egg
✓ Seaweed
✓ Green onions

Looks like: Ramen Bowl
```

```
[Edit if incorrect]

Portion size:
○ Small   ● Medium   ○ Large

[Analyze with Ayurveda →]
```

Step 5: Ayurvedic Analysis

Twin Message:

"Here's how this meal affects you:"

Analysis Card:

```
🍜 Ramen Bowl Analysis
──────────────────────────────────────

[Food Photo]

Agni Compatibility: 46/100
⚠️ Moderately Challenging

Why: This meal is heavy (noodles, broth) and
oily, requiring strong Agni to digest fully.
Your current Agni (68%) may struggle slightly.

Dosha Impact:
• Increases Pitta (salty, heating)
• Increases Kapha (heavy, oily)
• Neutral for Vata

Food Qualities:
• Hot (temperature & energetics)
• Heavy (hard to digest)
• Oily (greasy broth)
• Salty (high sodium)

Timing: 12:30 PM
✓ Good — Agni is strong at midday

Recommendations:
• Walk 10-15 min after eating
• Ginger tea to aid digestion
• Lighter dinner tonight
```

```
• Avoid similar heavy meals today

[See Full Details ↓]  [Save Meal]
```

3.5 Manual Meal Entry

Context: User chooses manual entry instead of photo.

Search Interface:

```
Add Food Items                                    [×]
──────────────────────────────────────────────────────

Search food:
┌──────────────────────────────────────────┐
│ 🔍 Start typing...                       │
└──────────────────────────────────────────┘

Recent:
• Oatmeal with berries
• Brown rice
• Grilled vegetables

Favorites:
⭐ Morning smoothie
⭐ Kitchari
⭐ Daal with rice

Common Ayurvedic Foods:
• Ghee
• Kitchari
• Mung dal
• Basmati rice
• Turmeric

[+ Add custom food]
```

Features:

- Real-time search with Ayurvedic food database
- Recent meals for quick re-logging
- Starred favorites
- Common Ayurvedic foods quick-add
- Custom food creation

3.6 Exploring the Constitutional Food Guide

Context: User scrolls to Constitutional Food Guide card.

Twin Message:

"Not sure what to eat? Let me show you foods that balance YOUR unique constitution..."

Collapsed Food Guide Card:

```
┌──────────────────────────────────────────────┐
│ 🥗 Your Constitutional Food Guide            │
│ ──────────────────────────────────────────── │
│                                              │
│ Based on your Pitta-Vata constitution:       │
│                                              │
│ Currently Recommended:                       │
│ ──────────────────────────────────────────── │
│                                              │
│ Your Vata is elevated, so favor:             │
│ • Warm, cooked foods                         │
│ • Grounding grains (rice, quinoa)            │
│ • Healthy fats (ghee, nuts)                  │
│ • Sweet, sour, salty tastes                  │
│                                              │
│ Foods to Support You Now:                    │
│ [Kitchari]  [Oatmeal]  [Avocado]             │
│                                              │
│          [See Full Guide ↓]                  │
└──────────────────────────────────────────────┘
```

Expanded Food Guide Contains:

1. **Constitution Overview**
   - Prakriti percentages
   - Current Vikriti state
   - Priority focus (which dosha to balance)
2. **Foods to Favor** (organized by category)
   - Grains
   - Vegetables
   - Proteins
   - Fruits
   - Fats
   - Spices
   - Tastes to favor
3. **Foods to Reduce**

    - Listed with explanations
    - Why they aggravate current state
4. **Secondary Considerations**
    - Baseline constitution needs
    - How to balance multiple considerations
5. **Seasonal Adjustments**
    - Current season impact
    - Seasonal food modifications
6. **Weather-Based Suggestions**
    - Today's weather impact
    - Specific adjustments
7. **Action Buttons**
    - [View Meal Plans →]
    - [See Recipes →]
    - [Shop Ingredients →]

3.7 Understanding Meal Timing

Context: User scrolls to Meal Timing card.

Twin Message:

"WHEN you eat is as important as WHAT you eat. Let me show you your optimal eating windows..."

Collapsed Meal Timing Card:

```
┌──────────────────────────────────────────────────────┐
│ ⏰ Your Optimal Meal Timing                          │
│ ──────────────────────────────────────────────────── │
│                                                      │
│ [Agni Curve Throughout Day]                          │
│                                                      │
│ Your Agni (digestive fire) peaks at noon —           │
│ make lunch your main meal!                           │
│                                                      │
│ Recommended Windows:                                 │
│ 🌅 Breakfast: 7-9 AM                                 │
│ ☀️ Lunch: 12-1 PM (Biggest meal)                     │
│ 🌙 Dinner: 6-7 PM (Light)                            │
│                                                      │
│ Meal Spacing: 4-6 hours apart                        │
│                                                      │
│            [See Full Details ↓]                      │
```

└──┘

Agni Curve Visualization:

- Line chart showing digestive strength
- X-axis: 6 AM to 10 PM
- Y-axis: Agni strength (0-100%)
- Peak at noon (12 PM)
- Meal windows highlighted
- Color: #F8C471 (gold/fire)

Expanded Meal Timing Card Contains:

1. **Detailed Agni Curve**
   - Larger chart with actual percentages
   - "6 AM: 45% → 12 PM: 90% (Peak) → 8 PM: 40%"
2. **Ayurvedic Principle**
   - "Your Agni follows the sun..."
   - Educational context
3. **Optimal Windows by Meal**
   - **Breakfast (7-9 AM)**
     - Agni: 60-70% (moderate)
     - What to eat: Light, warm, easy to digest
     - Size: Small to moderate
     - Examples: Oatmeal, fruit with nuts, warm smoothie
   - **Lunch (12-1 PM)** — MOST IMPORTANT
     - Agni: 85-90% (peak)
     - What to eat: Largest, heaviest meal
     - Size: Biggest meal of the day
     - Examples: Complete meal with protein, grains, vegetables
   - **Dinner (6-7 PM)**
     - Agni: 45-55% (declining)
     - What to eat: Light, easy to digest
     - Size: Smallest (30-50% of lunch)
     - Examples: Soup, kitchari, steamed vegetables
     - Note: Finish 3+ hours before bed
4. **Meal Spacing**
   - Ideal: 4-6 hours between meals
   - Why: Gives Agni time to fully digest
   - User's current pattern assessment
5. **Snacking Guidance**
   - Ideally avoid
   - If needed: Fresh fruit, soaked nuts, warm spiced milk
   - Avoid: Processed snacks, cold drinks
6. **Mindful Eating Practices**
   - Checklist of practices
   - Eat in calm, seated environment
   - No distractions
   - Chew thoroughly

    - Eat until 75% full
    - Short walk after meals
7. **Action Buttons**
    - [Set Meal Reminders →]
    - [Track Mindful Eating →]

3.8 Reviewing Weekly Trends

Context: User scrolls to Weekly Overview card.

Twin Message:

"Let's look at your eating patterns this week..."

Collapsed Weekly Card:

```
┌──────────────────────────────────────────────┐
│ 📊 This Week's Nutrition                      │
│ ──────────────────────────────────────────────│
│                                               │
│ [Mini Calendar View]                          │
│ Mon Tue Wed Thu Fri Sat Sun                   │
│  ✓   ✓   ✓   ✓   ✓   ○   ○                   │
│                                               │
│ Meals Logged: 15 of 21                        │
│ Consistency: 71%                              │
│                                               │
│ Average Agni Compatibility: 74/100            │
│ Most meals suit your digestion                │
│                                               │
│ Dosha Balance Trend:                          │
│ [Small trend chart]                           │
│ Slightly Pitta-heavy this week                │
│                                               │
│          [See Full Analysis ↓]                │
└──────────────────────────────────────────────┘
```

Expanded Weekly Card Contains:

1. **Meal Logging Consistency**
    - Visual calendar with checkmarks
    - "15 of 21 meals (71%)"
    - Encouragement for gaps
2. **Agni Compatibility Trends**
    - Bar chart by day
    - Most compatible meals (with scores)
    - Challenging meals (with scores)

3. **Dosha Balance Analysis**
    - Line chart showing daily dosha balance
    - Weekly averages with assessment
    - Which meals contributed to imbalances
4. **Eating Patterns**
    - Average meal times
    - Spacing assessment
    - Mindful eating score with breakdown
5. **Taste Balance**
    - Six tastes coverage analysis
    - Visual bars for each taste
    - Missing tastes highlighted
    - Foods to add for balance
6. **Recommendations for Next Week**
    - Specific, actionable items
    - Based on patterns observed
7. **Action Buttons**
    - [Create Weekly Meal Plan →]
    - [View Meal Ideas →]

3.9 Exploring Meal Plans & Recipes

Context: User wants structured meal guidance.

Meal Plans Card:

```
┌──────────────────────────────────────────────────────────┐
│ 📋 Personalized Meal Plans                               │
│ ──────────────────────────────────────────────────────── │
│                                                          │
│  ┌──────────────────────────────────────────────────┐    │
│  │ Vata-Balancing Week                              │    │
│  │ 7 days • 21 meals                                │    │
│  │ For: Elevated Vata                               │    │
│  │ [Preview meals →]                                │    │
│  └──────────────────────────────────────────────────┘    │
│                                                          │
│  ┌──────────────────────────────────────────────────┐    │
│  │ Pitta-Cooling Week                               │    │
│  │ 7 days • 21 meals                                │    │
│  │ For: Excess Pitta/Heat                           │    │
│  │ [Preview meals →]                                │    │
│  └──────────────────────────────────────────────────┘    │
│                                                          │
```

```
Create Custom Plan
Tailored to your current needs
[Start creating →]
```

Recipes Card:

```
👨‍🍳 Ayurvedic Recipes

Recommended for You:

[Recipe Image]
Kitchari (Healing Porridge)
⭐⭐⭐⭐⭐ Tri-doshic
Easy • 30 min • Serves 4
[View recipe →]

[Recipe Image]
Golden Milk (Turmeric Latte)
⭐⭐⭐⭐ Vata & Kapha balancing
Easy • 10 min • Serves 1
[View recipe →]

[Browse All Recipes →]
```

4. Technical Architecture

4.1 High-Level Architecture

```
PRESENTATION LAYER
```

```
│   │                        NutritionScreen
│   │
│   │   ┌──────────┐  ┌──────────┐  ┌──────────┐
┌──────────┐   │   │
│   │   │  Today's  │ │   Food    │ │   Meal    │ │   Weekly
│   │   │
│   │   │ Overview  │ │   Guide   │ │  Timing   │ │  Overview
│   │   │
│   │   └──────────┘  └──────────┘  └──────────┘
└──────────┘   │   │
│   │                      [+ LOG MEAL FAB]
│   │
│
└──────────────────────────────────────────────────────────────
─┘   │
└──────────────────────────────────────────────────────────────
──────────┘
                                  │
                                  ▼
┌──────────────────────────────────────────────────────────────
──────────┐
│                            STATE LAYER
│
│   ┌──────────────────┐   ┌──────────────────┐
┌──────────────────┐
│   │   Meals Store    │   │  Analysis Store  │   │  Nutrition
UI    │   │
│   │                  │   │                  │   │  Store
│   │
│   │ • Today's meals  │   │ • Dosha impact   │   │ • Expanded
cards │   │
│   │ • Meal history   │   │ • Agni scores    │   │ • Active
meal    │   │
│   │ • Favorites      │   │ • Patterns       │   │ • Log modal
│   │
│   │ • Recent foods   │   │ • Weekly trends  │   │ • Messages
│   └──────────────────┘   └──────────────────┘
└──────────────────┘   │
└──────────────────────────────────────────────────────────────
──────────┘
                                  │
                                  ▼
┌──────────────────────────────────────────────────────────────
──────────┐
```

```
│                          SERVICE LAYER
│
│   ┌──────────────────────┐
┌──────────────────────────────────────┐     │
│   │   Nutrition Agent    │  │      Food Analysis Service
│   │                      │  │
│   │   (Claude API)       │  │
│   │                      │  │
│   │                      │  │  • Agni compatibility calc
│   │                      │  │
│   │  • Food analysis     │  │  • Dosha impact analysis
│   │                      │  │
│   │  • Recommendations   │  │  • Quality assessment
│   │                      │  │
│   │  • Meal planning     │  │  • Pattern recognition
│   │                      │  │
│   │  • Insights          │  │  • Weekly aggregation
│   │                      │  │
│   └──────────────────────┘
└──────────────────────────────────────┘     │
│
│
│   ┌──────────────────────┐
┌──────────────────────────────────────┐     │
│   │   Food Recognition   │  │      Ayurvedic Food Database
│   │                      │  │
│   │   Service (AI)       │  │
│   │                      │  │
│   │                      │  │  • Food properties
│   │                      │  │
│   │  • Image analysis    │  │  • Dosha effects
│   │                      │  │
│   │  • Food detection    │  │  • Quality mappings
│   │                      │  │
│   │  • Portion estimate  │  │  • Recipe library
│   │                      │  │
│   └──────────────────────┘
└──────────────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
                                                   │
                                                   ▼
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│                          DATA LAYER
│
```

```
│   ┌──────────────────────┐
┌──────────────────────────────────────┐     │
│   │   Local Cache        │  │      Backend APIs
│   │                      │  │
│   │                      │  │
│   │ • Meal history       │  │  • Meal logging service
│   │                      │  │
│   │ • Food database      │  │  • Analysis service
│   │                      │  │
│   │ • Favorites          │  │  • Recipe service
│   │                      │  │
│   │ • Preferences        │  │  • Meal plan service
│   │                      │  │
│   └──────────────────────┘
└──────────────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────
──────────┘
```

4.2 Data Flow: Logging a Meal with Photo

```
User taps "+ LOG MEAL"
        │
        ▼
┌──────────────────────────┐
│ Show meal type           │ ── Breakfast/Lunch/Dinner/Snack
│ selection modal          │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Open camera              │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ User captures photo      │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Send to food             │ ── POST /nutrition/analyze-photo
│ recognition service      │    Returns: detected foods
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
```

```
┌──────────────────────────┐
│ Show recognition         │ ── User confirms or edits
│ results                  │    Selects portion size
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Send to Ayurvedic        │ ── POST /nutrition/log
│ analysis                 │    { foods, mealType, time,
photo }
└──────────────────────────┘
        │
        ├── Look up food properties in database
        ├── Calculate Agni compatibility
        ├── Determine dosha impact
        ├── Assess food qualities
        └── Generate recommendations
        │
        ▼
┌──────────────────────────┐
│ Claude generates         │ ── Personalized analysis text
│ analysis narrative       │    Contextual recommendations
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Display analysis         │ ── Full Ayurvedic meal analysis
│ card                     │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ User saves meal          │ ── Updates Today's Overview
└──────────────────────────┘    Refreshes dosha balance
                                Updates Agni compatibility
```

4.3 Data Flow: Generating Food Recommendations

```
User opens Constitutional Food Guide
        │
        ▼
┌──────────────────────────┐
│ Get user's Prakriti      │ ── From health profile
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Get current Vikriti      │ ── From Health section data
```

```
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Determine priority       │ ── Which dosha needs balancing?
│ dosha                    │
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Get current season       │ ── Seasonal adjustments
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Get today's weather      │ ── From Exposure section
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Query food database      │ ── Foods that balance priority
dosha
│ with all context         │    Adjusted for season & weather
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Claude generates         │ ── Personalized recommendations
│ guide narrative          │    With explanations
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Display Food Guide       │
│ card                     │
└──────────────────────────┘
```

4.4 Agni Compatibility Calculation

```
AGNI COMPATIBILITY CALCULATION:

Inputs:
- User's current Agni score (from Health section)
- Food's digestive difficulty
- Food's qualities (heavy/light, oily/dry, etc.)
- Meal timing (relative to Agni cycle)
- Previous meals today
```

```
Algorithm:

1. Base score from food difficulty:
   - Easy to digest: Start at 90
   - Moderate: Start at 70
   - Difficult: Start at 50
   - Very difficult: Start at 30

2. Adjust for user's Agni strength:
   - Strong Agni (80+): No penalty
   - Moderate Agni (60-79): -10 for difficult foods
   - Weak Agni (<60): -20 for difficult foods, -10 for
moderate

3. Adjust for timing:
   - Lunch (11am-2pm): +10 (peak Agni)
   - Breakfast (7-9am): +5
   - Early dinner (5-7pm): 0
   - Late dinner (after 8pm): -15

4. Adjust for meal load:
   - Previous heavy meal today: -10
   - Multiple heavy meals: -20

5. Adjust for qualities vs. constitution:
   - Food qualities match balancing needs: +5
   - Food qualities aggravate imbalance: -10

Final Score = Base + Adjustments (capped 0-100)
```

5. AI Integration & Prompting Strategy

5.1 The Role of Claude in Nutrition Section

Claude serves multiple functions:

Capability	Description
Food Recognition Analysis	Interpret AI-detected foods for Ayurvedic properties
Meal Analysis	Generate personalized analysis of logged meals
Recommendation Generation	Create specific, actionable eating guidance
Food Guide Creation	Build personalized constitutional food lists

Meal Planning	Generate dosha-balancing meal plans
Pattern Insights	Analyze weekly patterns and provide insights
Recipe Suggestions	Recommend recipes based on current needs

5.2 System Prompt for Nutrition Section

```
You are the Ayurvedic Digital Twin for AyuAware, serving as a
personal
nutrition guide. Your role is to help users understand how
their food
choices affect their unique constitution and provide
actionable guidance.

## YOUR ROLE IN THIS SECTION

You are an Ayurvedic nutritionist. You don't count calories —
you evaluate
food through the lens of qualities, digestibility, and
constitutional impact.
Every meal is an opportunity to balance or imbalance the
doshas.

## AYURVEDIC NUTRITION PRINCIPLES

Food Qualities (Gunas):
- Hot/Cold (temperature AND energetics)
- Heavy/Light (digestibility)
- Oily/Dry (moisture content)
- Smooth/Rough (texture)
- Dense/Flowing (consistency)

Six Tastes:
- Sweet (nourishing, grounding) - pacifies Vata, Pitta
- Sour (stimulating, heating) - pacifies Vata
- Salty (moistening, heating) - pacifies Vata
- Pungent (stimulating, drying) - pacifies Kapha
- Bitter (cooling, detoxifying) - pacifies Pitta, Kapha
- Astringent (drying, cooling) - pacifies Pitta, Kapha

Food Categories by Dosha:

VATA-BALANCING: Warm, moist, grounding
- Favor: Cooked foods, healthy fats, sweet/sour/salty tastes
- Reduce: Raw foods, dry foods, bitter/astringent tastes
```

PITTA-BALANCING: Cool, moderate, not too oily
- Favor: Cooling foods, sweet/bitter/astringent tastes
- Reduce: Spicy, sour, salty, fried foods

KAPHA-BALANCING: Light, warm, stimulating
- Favor: Light foods, pungent/bitter/astringent tastes
- Reduce: Heavy, oily, sweet, cold foods

AGNI (DIGESTIVE FIRE) PRINCIPLES

- Agni follows the sun: strongest at midday
- Heavy meals need strong Agni to digest
- Weak Agni creates Ama (toxins) regardless of food quality
- Lunch should be the largest meal
- Dinner should be light and early

MEAL TIMING PRINCIPLES

- Breakfast: 7-9 AM (light, warm)
- Lunch: 12-1 PM (largest meal, peak Agni)
- Dinner: 6-7 PM (light, easily digestible)
- Spacing: 4-6 hours between meals
- Finish eating: 3+ hours before bed

MINDFUL EATING PRINCIPLES

- Eat in calm environment
- No distractions (phone, TV, work)
- Chew thoroughly (20-30 times)
- Eat until 75% full
- Express gratitude
- Walk briefly after meals

WHEN ANALYZING A MEAL

1. Identify the foods and their qualities
2. Assess Agni compatibility (can they digest this?)
3. Determine dosha impact (does it balance or aggravate?)
4. Evaluate timing (is this the right time for this food?)
5. Generate specific recommendations

COMMUNICATION APPROACH

When analyzing meals:

```
- Be specific about which qualities affect them
- Explain WHY, not just what
- Always end with actionable recommendations
- Don't be judgmental about "bad" food choices
- Frame as learning opportunities

When recommending foods:
- Prioritize their primary imbalance
- Consider season and weather
- Be specific (not just "eat warm foods" but "try oatmeal
with ghee")
- Include multiple options

## WHAT NOT TO DO

- Don't count calories or macros (unless asked)
- Don't use guilt or shame about food choices
- Don't recommend dramatic dietary changes
- Don't ignore their stated preferences
- Don't contradict established Ayurvedic principles
```

5.3 Context Building for Nutrition Section

Every Claude call includes:

```
CONTEXT PACKAGE STRUCTURE:

## User Profile
- Name: {user_name}
- Prakriti: {prakriti_percentages}
- Primary constitution: {constitution_type}
- Current Vikriti: {vikriti_percentages}
- Current imbalances: {imbalance_list}
- Current Agni score: {agni_score}%
- Agni type: {agni_type}

## Today's Nutrition
- Meals logged: {meal_count}
- Meal details: {meal_list_with_times}
- Running dosha balance: {dosha_percentages}
- Running Agni compatibility: {average_score}
- Eating patterns: {mindful_eating_scores}

## Current Context
- Current time: {current_time}
- Current Agni strength: {agni_at_this_time}%
```

```
- Today's weather: {weather_summary}
- Season: {current_season}

## Meal Being Analyzed (if applicable)
- Foods detected: {food_list}
- Portion size: {portion}
- Meal type: {breakfast/lunch/dinner/snack}
- Time eaten: {meal_time}

## User Preferences
- Dietary restrictions: {restrictions}
- Food preferences: {preferences}
- Detail level: {brief/balanced/detailed}
```

5.4 Prompt Templates by Scenario

Meal Analysis:

```
Analyze this meal from an Ayurvedic perspective.

{CONTEXT_PACKAGE}

Meal to Analyze:
- Foods: {food_list}
- Portion: {portion_size}
- Meal type: {meal_type}
- Time: {meal_time}

Generate an analysis that includes:

1. **Agni Compatibility Score** (0-100)
   - Score and category (Excellent/Good/Moderate/Challenging)
   - Why this meal is easy or difficult to digest
   - How their current Agni handles it

2. **Dosha Impact**
   - Which doshas does this increase/decrease?
   - Is this beneficial given current imbalances?

3. **Food Qualities Assessment**
   - Hot/cold, heavy/light, oily/dry
   - How these qualities affect them

4. **Timing Assessment**
   - Is this the right time for this food?
   - How does current Agni strength affect digestion?
```

5. **Recommendations** (4-5 specific items)
   - What to do after this meal
   - What to adjust for remaining meals today
   - Any digestive support needed

Tone: Informative, supportive, never judgmental
Length: Moderate (can be expanded for full details)

Food Guide Generation:

Generate a personalized food guide for this user.

{CONTEXT_PACKAGE}

Generate comprehensive guidance including:

1. **Priority Focus**
   - Which dosha needs balancing most
   - Why (based on Vikriti vs Prakriti)

2. **Foods to Favor** (organized by category)
   - Grains
   - Vegetables
   - Proteins
   - Fruits
   - Fats/Oils
   - Spices
   - Tastes to emphasize

3. **Foods to Reduce**
   - Specific foods that aggravate
   - Why each is problematic

4. **Secondary Considerations**
   - Baseline constitution needs
   - How to balance multiple factors

5. **Seasonal Adjustments**
   - Current season: {season}
   - Modifications for this time of year

6. **Weather Adjustments**
   - Today: {weather}
   - Specific adjustments

Make recommendations specific and actionable.
Include reasoning for major recommendations.

Weekly Analysis:

```
Analyze this user's nutrition patterns for the week.

{CONTEXT_PACKAGE}

Weekly Data:
- Meals logged: {meals_with_scores}
- Daily dosha balances: {daily_dosha_data}
- Meal timing patterns: {timing_data}
- Mindful eating scores: {mindful_data}
- Taste distribution: {taste_data}

Generate analysis including:

1. **Logging Consistency**
   - Assessment and encouragement

2. **Agni Compatibility Trends**
   - Average score
   - Best and worst meals
   - Patterns observed

3. **Dosha Balance Trends**
   - Weekly trend
   - Which meals contributed to imbalances
   - Overall assessment

4. **Eating Pattern Analysis**
   - Timing consistency
   - Spacing assessment
   - Mindful eating score

5. **Taste Balance**
   - Coverage of six tastes
   - What's missing
   - Foods to add

6. **Recommendations for Next Week**
   - 4-5 specific, actionable items
   - Based on observed patterns

Tone: Encouraging, specific, forward-looking
```

Meal Plan Generation:

```
Create a personalized meal plan for this user.

{CONTEXT_PACKAGE}

Plan Parameters:
- Duration: {days} days
- Focus: {dosha_to_balance}
- Restrictions: {dietary_restrictions}
- Preferences: {food_preferences}

Generate a complete meal plan with:

For each day:
- Breakfast (with time)
- Lunch (largest meal, with time)
- Dinner (light, with time)

For each meal:
- Specific foods/dishes
- Portion guidance
- Why this meal helps

Include:
- Variety across the week
- Practical, easy-to-prepare options
- Seasonal appropriateness
- Shopping list suggestions
- Preparation tips

Make it achievable, not overwhelming.
```

5.5 Food Database Integration

Claude references an Ayurvedic food database with:

```
FOOD DATABASE STRUCTURE:

Food:
  - name: string
  - aliases: string[] (common names)
  - category: 'grain' | 'vegetable' | 'fruit' | 'protein' |
              'dairy' | 'fat' | 'spice' | 'beverage' |
'prepared'
  - qualities:
      - thermal: 'heating' | 'cooling' | 'neutral'
      - weight: 'heavy' | 'light' | 'moderate'
```

```
      - moisture: 'oily' | 'dry' | 'balanced'
      - texture: 'smooth' | 'rough' | 'varied'
  - tastes: string[] (sweet, sour, salty, pungent, bitter,
astringent)
  - doshaEffect:
      - vata: 'increases' | 'decreases' | 'neutral'
      - pitta: 'increases' | 'decreases' | 'neutral'
      - kapha: 'increases' | 'decreases' | 'neutral'
  - digestibility: 'easy' | 'moderate' | 'difficult'
  - bestSeasons: string[]
  - notes: string (preparation tips, combinations to avoid)
  - sattvicRajasicTamasic: 'sattvic' | 'rajasic' | 'tamasic'

Example:
{
  name: "Oatmeal",
  aliases: ["oats", "porridge"],
  category: "grain",
  qualities: {
    thermal: "heating",
    weight: "moderate",
    moisture: "balanced",
    texture: "smooth"
  },
  tastes: ["sweet"],
  doshaEffect: {
    vata: "decreases",
    pitta: "neutral",
    kapha: "increases"
  },
  digestibility: "easy",
  bestSeasons: ["fall", "winter"],
  notes: "Best cooked with ghee. Avoid with cold milk.",
  sattvicRajasicTamasic: "sattvic"
}
```

6. Data Models & State Management

6.1 Core Data Types

Meal:

```
Meal:
  - id: string
  - userId: string
```

```
  - date: Date
  - mealType: 'breakfast' | 'lunch' | 'dinner' | 'snack'
  - time: string (HH:mm)
  - foods: FoodItem[]
  - photoUrl: string | null
  - analysis: MealAnalysis
  - mindfulEating: MindfulEatingData | null
  - createdAt: Date
  - updatedAt: Date

FoodItem:
  - foodId: string
  - name: string
  - portion: 'small' | 'medium' | 'large'
  - customNotes: string | null

MealAnalysis:
  - agniCompatibility: number (0-100)
  - agniCategory: 'excellent' | 'good' | 'moderate' |
'challenging' | 'avoid'
  - doshaImpact: DoshaImpact
  - qualities: FoodQualities
  - timingAssessment: TimingAssessment
  - recommendations: string[]
  - fullAnalysis: string (Claude-generated narrative)

DoshaImpact:
  - vata: number (-100 to +100, negative = decreases)
  - pitta: number
  - kapha: number
  - primaryEffect: string (e.g., "Pitta-aggravating")

FoodQualities:
  - thermal: 'heating' | 'cooling' | 'neutral'
  - weight: 'heavy' | 'light' | 'moderate'
  - moisture: 'oily' | 'dry' | 'balanced'
  - dominantTastes: string[]

TimingAssessment:
  - appropriateness: 'optimal' | 'good' | 'suboptimal' |
'poor'
  - agniAtTime: number (0-100)
  - explanation: string

MindfulEatingData:
```

```
  - environment: 'calm' | 'rushed' | 'moderate'
  - distraction: 'none' | 'some' | 'significant'
  - paceAssessment: 'slow' | 'moderate' | 'fast'
```

Daily Nutrition Summary:

```
DailyNutritionSummary:
  - date: Date
  - meals: Meal[]
  - mealsLogged: number
  - mealsExpected: number (usually 3)
  - overallDoshaBalance: DoshaBalance
  - averageAgniCompatibility: number
  - eatingPatterns: EatingPatterns
  - recommendations: string[]
  - insight: string

DoshaBalance:
  - vata: number (percentage)
  - pitta: number
  - kapha: number
  - assessment: string (e.g., "Slightly Pitta-heavy")

EatingPatterns:
  - mealTiming: TimingPattern
  - mealSpacing: SpacingPattern
  - mindfulEatingScore: number (0-100)

TimingPattern:
  - breakfastTime: string | null
  - lunchTime: string | null
  - dinnerTime: string | null
  - assessment: 'good' | 'moderate' | 'needs_improvement'

SpacingPattern:
  - averageSpacing: number (hours)
  - assessment: 'optimal' | 'good' | 'too_close' | 'too_far'
```

Weekly Nutrition Summary:

```
WeeklyNutritionSummary:
  - weekStart: Date
  - weekEnd: Date
  - daysWithLogs: number
  - totalMealsLogged: number
  - totalMealsExpected: number
  - consistency: number (percentage)
  - averageAgniCompatibility: number
```

```
  - agniTrend: DailyAgniScore[]
  - doshaBalance: WeeklyDoshaBalance
  - tasteBalance: TasteBalance
  - eatingPatterns: WeeklyEatingPatterns
  - bestMeals: MealHighlight[]
  - challengingMeals: MealHighlight[]
  - recommendations: string[]

WeeklyDoshaBalance:
  - dailyValues: DailyDoshaValues[]
  - weeklyAverage: DoshaBalance
  - trend: 'improving' | 'stable' | 'worsening'
  - contributingFactors: string[]

TasteBalance:
  - sweet: number (0-100)
  - sour: number
  - salty: number
  - pungent: number
  - bitter: number
  - astringent: number
  - missingTastes: string[]
  - recommendations: string[]

MealHighlight:
  - mealId: string
  - name: string
  - agniScore: number
  - date: Date
```

Food Guide:

```
ConstitutionalFoodGuide:
  - userId: string
  - generatedAt: Date
  - prakriti: DoshaPercentages
  - vikriti: DoshaPercentages
  - priorityDosha: 'vata' | 'pitta' | 'kapha'
  - foodsToFavor: FoodRecommendations
  - foodsToReduce: FoodRecommendations
  - tastesToFavor: string[]
  - tastesToReduce: string[]
  - seasonalAdjustments: string[]
  - weatherAdjustments: string[]
  - fullGuide: string (Claude-generated narrative)
```

```
FoodRecommendations:
  - grains: FoodWithReason[]
  - vegetables: FoodWithReason[]
  - fruits: FoodWithReason[]
  - proteins: FoodWithReason[]
  - fats: FoodWithReason[]
  - spices: FoodWithReason[]

FoodWithReason:
  - food: string
  - reason: string
```

Meal Plan:

```
MealPlan:
  - id: string
  - name: string
  - duration: number (days)
  - focus: 'vata_balancing' | 'pitta_balancing' |
'kapha_balancing' | 'custom'
  - days: MealPlanDay[]
  - shoppingList: ShoppingItem[]
  - createdAt: Date
  - isActive: boolean

MealPlanDay:
  - dayNumber: number
  - date: Date | null (if active)
  - breakfast: PlannedMeal
  - lunch: PlannedMeal
  - dinner: PlannedMeal

PlannedMeal:
  - name: string
  - description: string
  - recipeId: string | null
  - prepTime: number (minutes)
  - doshaEffect: string
  - notes: string

ShoppingItem:
  - name: string
  - quantity: string
  - category: string
  - productId: string | null (for shop link)
```

Recipe:

```
Recipe:
  - id: string
  - name: string
  - description: string
  - doshaEffect: string (e.g., "Tri-doshic", "Vata-
balancing")
  - difficulty: 'easy' | 'moderate' | 'advanced'
  - prepTime: number (minutes)
  - cookTime: number (minutes)
  - servings: number
  - ingredients: Ingredient[]
  - instructions: string[]
  - tips: string[]
  - imageUrl: string
  - rating: number
  - tags: string[]
```

6.2 State Structure

```
NutritionState:
  # Visit Context
  - isFirstVisit: boolean
  - lastVisit: Date | null

  # Today's Data
  - todaysSummary: DailyNutritionSummary | null
  - isLoadingToday: boolean
  - todayError: Error | null

  # Meal Logging
  - activeMeal: Meal | null
  - isLogging: boolean
  - isAnalyzing: boolean
  - recognitionResults: FoodRecognitionResult | null

  # Food Guide
  - foodGuide: ConstitutionalFoodGuide | null
  - isLoadingGuide: boolean

  # Weekly Data
  - weeklySummary: WeeklyNutritionSummary | null
  - isLoadingWeekly: boolean

  # Meal Plans & Recipes
  - mealPlans: MealPlan[]
  - activeMealPlan: MealPlan | null
```

```
  - recipes: Recipe[]
  - isLoadingRecipes: boolean

  # User Preferences
  - favorites: string[] (food IDs)
  - recentFoods: FoodItem[]
  - dietaryRestrictions: string[]

  # UI State
  - expandedCards: string[]
  - logModalOpen: boolean
  - selectedMealType: MealType | null
  - scrollPosition: number

  # Chat
  - messages: NutritionMessage[]
  - isTyping: boolean
```

6.3 State Management Approach

Zustand Stores:

```
Stores:

1. mealsStore
   - Today's meals
   - Meal history
   - Logging operations
   - CRUD operations

2. analysisStore
   - Meal analysis results
   - Daily summaries
   - Weekly summaries
   - Pattern calculations

3. foodGuideStore
   - Constitutional recommendations
   - Seasonal adjustments
   - Food database queries

4. recipesStore
   - Recipe library
   - Meal plans
   - Favorites
```

```
5. nutritionUIStore
   - First visit flag
   - Expanded cards
   - Modal states
   - Messages
```

7. Component Requirements

7.1 Overview Components

NutritionOverviewCard

Displays today's nutrition summary:

- Meal status row (logged/pending)
- Dosha balance chart
- Agni compatibility score
- Eating pattern insights
- Expand button

States:

- Collapsed: Summary view
- Expanded: Full day detail with meal cards

MealStatusRow

- Horizontal layout: Breakfast | Lunch | Dinner | (Snack)
- Checkmark for logged meals
- Circle for pending meals
- Tappable to log or view

DoshaBalanceChart

- Small tri-color donut
- Size: 60x60px (collapsed), 120x120px (expanded)
- Animated transitions
- Legend with percentages

7.2 Meal Logging Components

FloatingActionButton

- Position: Bottom-right, sticky
- Size: 56x56px
- Background: #4A9B8E
- Icon: + (24x24px, white)
- Label: "LOG MEAL" (on scroll up)
- Always visible during scroll

MealLogModal

- Bottom sheet style
- Sections: Meal type selector, logging methods
- Methods: Photo, Manual, Recent & Favorites

CameraInterface

- Full-screen viewfinder
- Guidelines overlay
- Tips text
- Capture/Cancel/Gallery buttons

FoodRecognitionResults

- Photo preview
- Detected foods list with checkmarks
- Edit option
- Portion selector (Small/Medium/Large)
- "Analyze with Ayurveda" button

ManualEntrySearch

- Search input
- Recent foods list
- Favorites list (starred)
- Ayurvedic foods suggestions
- Custom food creation

7.3 Analysis Components

MealAnalysisCard

Full Ayurvedic analysis display:

- Food photo (if available)
- Agni Compatibility score with color
- Dosha impact assessment
- Food qualities list
- Timing assessment
- Recommendations list
- Save button

AgniCompatibilityBadge

- Score (0-100)
- Color-coded background
- Category label
- Size: Inline or prominent

Color Coding:

- Excellent (85-100): #4A9B8E (green)
- Good (70-84): #7AB8AC (light teal)
- Moderate (55-69): #F8C471 (gold)
- Challenging (40-54): #E89B7F (coral)
- Avoid (<40): #D64545 (red)

DoshaImpactDisplay

- Three rows (Vata, Pitta, Kapha)
- Increase/decrease/neutral indicators
- Color-coded by effect
- Primary effect highlighted

7.4 Food Guide Components

ConstitutionalFoodGuideCard

- Constitution summary
- Priority focus
- Foods to favor (by category)
- Foods to reduce
- Seasonal/weather adjustments
- Expandable sections

FoodCategoryList

- Category header
- Food items with icons
- Brief reason per food
- Tappable for more info

SeasonalBadge

- Current season indicator
- Adjustment recommendations

7.5 Meal Timing Components

MealTimingCard

- Agni curve visualization
- Optimal windows
- Meal spacing guidance
- Mindful eating checklist

AgniCurveChart

- Line chart showing digestive strength
- X-axis: 6 AM to 10 PM
- Y-axis: 0-100%
- Peak highlighted at noon
- Meal windows shaded
- Color: #F8C471 (gold/fire)
- Size: Full width, 120px height

MealWindowIndicator

- Time range display
- Agni strength at that time
- Recommended meal size
- Ideal foods summary

7.6 Weekly Components

WeeklyNutritionCard

- Calendar view with logging status
- Consistency percentage
- Average Agni compatibility
- Dosha trend chart
- Taste balance bars
- Recommendations

WeekCalendarView

- 7 columns (Mon-Sun)
- Checkmark/circle for each day
- Tappable for day detail

TasteBalanceBars

- 6 horizontal bars (one per taste)
- Fill percentage
- Low/good indicators
- Foods to add for missing tastes

7.7 Meal Plans & Recipes Components

MealPlanCard

- Plan name and duration
- Focus (which dosha)
- Preview button
- Start/activate button

RecipeCard

- Recipe image (120x120px)
- Name
- Dosha effect badge
- Difficulty and time
- Rating
- View button

RecipeDetailView

- Full image
- Ingredients list
- Step-by-step instructions
- Tips
- Shop ingredients button

7.8 Animation Requirements

Animation	Duration	Easing	Trigger
FAB appear/hide	200ms	ease-out	Scroll direction
Modal slide up	300ms	ease-out	Open modal
Modal slide down	250ms	ease-in	Close modal
Photo capture	100ms	ease-out	Shutter
Analysis loading	Continuous	linear	During analysis
Score reveal	600ms	ease-out	Analysis complete
Chart fill	800ms	ease-out	Card appears

Card expand	400ms	ease-out	Tap expand
Checkmark appear	200ms	ease-out	Meal logged

8. API Requirements

8.1 REST Endpoints

Get Today's Nutrition:

```
GET /api/v1/nutrition/today

Response:
{
  "date": "2025-01-31",
  "meals": [
    {
      "id": "meal_001",
      "mealType": "breakfast",
      "time": "07:45",
      "foods": [...],
      "analysis": {
        "agniCompatibility": 85,
        "doshaImpact": {...},
        ...
      }
    },
    ...
  ],
  "summary": {
    "mealsLogged": 2,
    "mealsExpected": 3,
    "overallDoshaBalance": {...},
    "averageAgniCompatibility": 78,
    "eatingPatterns": {...},
    "recommendations": [...],
    "dinnerSuggestions": [...]
  }
}
```

Log a Meal:

```
POST /api/v1/nutrition/log

Request:
{
```

```
  "mealType": "lunch",
  "time": "12:30",
  "foods": [
    { "foodId": "food_123", "name": "Ramen", "portion":
"medium" }
  ],
  "photoUrl": "https://...",
  "mindfulEating": {
    "environment": "rushed",
    "distraction": "some"
  }
}

Response:
{
  "meal": { ... },
  "analysis": {
    "agniCompatibility": 46,
    "agniCategory": "moderate",
    "doshaImpact": {
      "vata": 0,
      "pitta": 35,
      "kapha": 25,
      "primaryEffect": "Pitta-aggravating"
    },
    "qualities": {...},
    "timingAssessment": {...},
    "recommendations": [...],
    "fullAnalysis": "Your lunch of ramen..."
  }
}
```

Analyze Photo:

```
POST /api/v1/nutrition/analyze-photo

Request:
{
  "photo": "<base64 or URL>"
}

Response:
{
  "detectedFoods": [
    { "name": "Ramen noodles", "confidence": 0.95 },
    { "name": "Boiled egg", "confidence": 0.92 },
```

```
    ...
  ],
  "suggestedMealName": "Ramen Bowl",
  "requiresConfirmation": true
}
```

Get Food Guide:

```
GET /api/v1/nutrition/food-guide

Response:
{
  "prakriti": {...},
  "vikriti": {...},
  "priorityDosha": "vata",
  "foodsToFavor": {
    "grains": [...],
    "vegetables": [...],
    ...
  },
  "foodsToReduce": {...},
  "tastesToFavor": ["sweet", "sour", "salty"],
  "seasonalAdjustments": [...],
  "weatherAdjustments": [...],
  "fullGuide": "Based on your Pitta-Vata constitution..."
}
```

Get Weekly Summary:

```
GET /api/v1/nutrition/weekly

Response:
{
  "weekStart": "2025-01-25",
  "weekEnd": "2025-01-31",
  "daysWithLogs": 5,
  "totalMealsLogged": 15,
  "consistency": 71,
  "averageAgniCompatibility": 74,
  "agniTrend": [...],
  "doshaBalance": {...},
  "tasteBalance": {...},
  "eatingPatterns": {...},
  "bestMeals": [...],
  "challengingMeals": [...],
  "recommendations": [...]
}
```

Get Meal Plans:

```
GET /api/v1/nutrition/meal-plans

Response:
{
  "plans": [
    {
      "id": "plan_001",
      "name": "Vata-Balancing Week",
      "duration": 7,
      "focus": "vata_balancing",
      "preview": "Warm, grounding meals..."
    },
    ...
  ]
}
```

Create Custom Meal Plan:

```
POST /api/v1/nutrition/create-meal-plan

Request:
{
  "duration": 7,
  "focus": "custom",
  "doshaFocus": "vata",
  "restrictions": ["vegetarian"],
  "preferences": ["simple recipes"]
}

Response:
{
  "plan": {
    "id": "plan_custom_001",
    "name": "Your Custom Plan",
    "days": [...],
    "shoppingList": [...]
  }
}
```

Get Recipes:

```
GET /api/v1/nutrition/recipes?dosha=vata&difficulty=easy

Response:
{
  "recipes": [
    {
      "id": "recipe_001",
```

```
      "name": "Kitchari",
      "doshaEffect": "Tri-doshic",
      "difficulty": "easy",
      "prepTime": 10,
      "cookTime": 30,
      "imageUrl": "...",
      "rating": 4.8
    },
    ...
  ]
}
```

Search Foods:

```
GET /api/v1/nutrition/foods/search?q=oatmeal

Response:
{
  "results": [
    {
      "id": "food_oatmeal",
      "name": "Oatmeal",
      "category": "grain",
      "doshaEffect": {
        "vata": "decreases",
        "pitta": "neutral",
        "kapha": "increases"
      },
      "digestibility": "easy"
    },
    ...
  ]
}
```

8.2 WebSocket Events (Optional)

```
WebSocket Events:

meal.logged
{
  "mealId": "meal_001",
  "mealType": "lunch",
  "timestamp": "2025-01-31T12:30:00Z"
}

analysis.complete
{
```

```
  "mealId": "meal_001",
  "analysis": {...}
}

daily.updated
{
  "date": "2025-01-31",
  "summary": {...}
}
```

9. Success Metrics

9.1 Engagement Metrics

Metric	Target	Measurement
Section visits per week	> 5	Users visiting Nutrition weekly
Meals logged per day	> 2	Average meals logged
Logging consistency	> 60%	Users logging 60%+ of meals
Photo vs manual logging	Track	Ratio of logging methods
Return rate	> 50%	Users returning within 7 days

9.2 Feature Usage Metrics

Metric	Target	Measurement
Photo logging usage	> 40%	Meals logged with photo
Food guide views	> 30%	Users who view food guide
Weekly review views	> 40%	Users who check weekly summary
Recipe views	> 20%	Users who view recipes
Meal plan creation	> 10%	Users who create meal plans

9.3 Action Metrics

Metric	Target	Measurement
Post-meal actions taken	> 30%	Users who follow recommendations
Recipe try rate	> 15%	Users who log recipe meals
Meal plan adherence	> 40%	Meals matching active plan

Shop click-through	> 10%	Users clicking shop links

9.4 Health Outcome Metrics

Metric	Target	Measurement
Agni compatibility trend	Improving	Week-over-week average
Dosha balance trend	Improving	Alignment with recommendations
Mindful eating score	Improving	Week-over-week average
Taste balance	Improving	Coverage of six tastes

9.5 Technical Metrics

Metric	Target	Measurement
Photo analysis time	< 3s	Time from capture to recognition
Meal analysis time	< 2s	Time to generate full analysis
Section load time	< 2s	Time to first content
Food search response	< 500ms	Search results return time
Error rate	< 2%	Failed logs or analyses

10. Implementation Considerations

10.1 Food Recognition Integration

AI Model Options:

- Google Cloud Vision API
- AWS Rekognition
- Custom trained model
- Third-party food recognition API

Recognition Workflow:

1. Capture/upload image
2. Send to recognition service
3. Get list of detected foods with confidence scores
4. Map to Ayurvedic food database
5. Allow user correction
6. Store mapping for improvement

Handling Recognition Errors:

- Allow manual override

- Suggest similar foods
- "Can't identify? Enter manually" option
- Learn from corrections

10.2 Offline Handling

When Offline:

- Allow meal logging (queue for sync)
- Show cached food guide
- Display previous analyses
- Manual entry works fully
- Photo capture saved for later analysis

When Back Online:

- Sync queued meals
- Run pending analyses
- Update recommendations
- Refresh food guide if stale

10.3 Performance Optimization

Photo Handling:

- Compress images before upload
- Progressive JPEG loading
- Cache analyzed photos
- Thumbnail generation

Data Loading:

- Load today's summary first
- Lazy-load weekly data
- Cache food database locally
- Prefetch likely recipes

10.4 Accessibility Requirements

Screen Reader:

- Meal status announced
- Analysis scores read with context
- Food lists navigable
- Charts have text alternatives

Motor Accessibility:

- All tap targets ≥ 44x44px
- FAB always reachable
- Swipe to dismiss modals
- Voice input for food search

Visual Accessibility:

- WCAG AA contrast
- Don't rely on color alone for scores
- Support Dynamic Type
- High contrast mode support

10.5 Analytics Events

Event	Properties	Purpose
`nutrition_opened`	source, is_first_visit	Track entry
`meal_logged`	meal_type, method, food_count	Logging behavior
`photo_analyzed`	foods_detected, accuracy	AI performance
`analysis_viewed`	meal_id, agni_score	Engagement
`food_guide_viewed`	priority_dosha	Guide usage
`weekly_viewed`	consistency_score	Review behavior
`recipe_viewed`	recipe_id, dosha_effect	Recipe interest
`meal_plan_created`	duration, focus	Plan creation
`shop_clicked`	product_category	Commerce interest
`recommendation_followed`	rec_type	Action taking

10.6 Testing Priorities

Unit Tests:

- Agni compatibility calculation
- Dosha impact calculation
- Meal timing assessment
- Taste balance calculation

Integration Tests:

- Photo → Recognition → Analysis flow
- Manual entry → Analysis flow
- Food guide generation
- Weekly summary calculation

E2E Tests:

- First-time user education flow
- Complete meal logging with photo
- Weekly review flow
- Meal plan creation and activation

Manual QA:

- Photo recognition accuracy
- Analysis relevance and personalization
- Food guide accuracy for different constitutions
- Recipe filtering correctness
- UI responsiveness during analysis

Appendix A: Agni Compatibility Reference

Agni Strength by Time

Time	Agni Strength	Best For
6-7 AM	50%	Light breakfast
7-9 AM	60-70%	Moderate breakfast
9-11 AM	75%	Substantial breakfast
11 AM-1 PM	85-90%	Heaviest meal (lunch)
1-3 PM	80%	Late lunch OK
3-5 PM	60%	Light snack only
5-7 PM	50-55%	Light dinner
7-9 PM	40%	Very light only
After 9 PM	<35%	Avoid eating

Food Digestibility Categories

Category	Examples	Agni Required
Easy	Fruit, rice, mung dal	40%
Moderate	Vegetables, quinoa, chicken	55%
Difficult	Cheese, beans, red meat	70%
Very Difficult	Fried foods, raw salads, heavy desserts	80%

Appendix B: Six Tastes Reference

Taste Effects on Doshas

Taste	Vata	Pitta	Kapha
Sweet	↓ Decreases	↓ Decreases	↑ Increases
Sour	↓ Decreases	↑ Increases	↑ Increases
Salty	↓ Decreases	↑ Increases	↑ Increases
Pungent	↑ Increases	↑ Increases	↓ Decreases
Bitter	↑ Increases	↓ Decreases	↓ Decreases
Astringent	↑ Increases	↓ Decreases	↓ Decreases

Taste Sources

Taste	Common Foods
Sweet	Grains, dairy, sweet fruits, root vegetables
Sour	Citrus, fermented foods, yogurt, vinegar
Salty	Salt, seaweed, tamari, celery
Pungent	Ginger, garlic, onion, hot peppers, black pepper
Bitter	Leafy greens, turmeric, coffee, dark chocolate
Astringent	Beans, lentils, pomegranate, unripe banana

Appendix C: Food Quality Reference

Thermal Quality

Heating	Cooling	Neutral
Ginger	Cucumber	Rice
Garlic	Mint	Mung dal
Onion	Coconut	Ghee
Peppers	Cilantro	Milk
Tomato	Melon	Oats

Weight Quality

Heavy	Light	Moderate
Cheese	Leafy greens	Most grains
Red meat	Apples	Chicken
Wheat	Barley	Eggs
Yogurt	Corn	Fish
Nuts	Millet	Most vegetables

Moisture Quality

Oily	Dry	Balanced

Avocado	Crackers	Rice
Nuts	Popcorn	Vegetables
Ghee	Dried fruit	Fruit
Coconut	Toast	Dal
Oils	Beans	Oatmeal

Appendix D: Sample Meal Analyses

Example 1: Excellent Compatibility

Meal: Kitchari with ghee (lunch at 12:30 PM)

```
Agni Compatibility: 95/100 ✓ Excellent

Why: Kitchari is the most easily digestible meal in
Ayurveda. The combination of rice and mung dal is
complete protein while remaining light. Ghee enhances
absorption and soothes digestion.

Dosha Impact:
• Vata: Decreases (grounding, warm)
• Pitta: Decreases (cooling qualities of ghee)
• Kapha: Neutral (light enough)

Timing: Perfect — eaten at peak Agni

Recommendations:
• Ideal recovery meal
• Can eat slightly larger portion
• No digestive support needed
```

Example 2: Challenging Compatibility

Meal: Cheese pizza with salad (dinner at 8:30 PM)

```
Agni Compatibility: 32/100 ⚠️ Challenging

Why: This meal combines heavy cheese with raw vegetables,
eaten late when Agni is only at 35%. The combination of
hard-to-digest cheese with cold, rough salad creates
opposing demands on weakened digestion.

Dosha Impact:
```

- Vata: Increases (cold salad, late timing)
- Pitta: Neutral
- Kapha: Increases significantly (cheese, heavy)

Timing: Poor — Agni too weak for this meal

Recommendations:
- Take a short walk (10-15 min)
- Sip warm ginger tea
- Expect sluggish morning
- Tomorrow: light breakfast, larger lunch
- Avoid similar late/heavy meals

End of Document